Unit 3 2017.pdf

SECTION B

Answer **all** *questions*.

Read through the following article carefully.

Lasers in Outer Space

Paragraph

Einstein's theory of stimulated emission, published in 1916, laid the foundation for lasers but it took humans another 44 years before the first successful ruby laser was produced. However, seven years beforehand a maser had been produced. This is similar to a laser but involves microwaves instead of light – **m**icrowave **a**mplification by **s**timulated **e**mission of **r**adiation. 1

It turns out that masers are easier to build than lasers because the lifetime of metastable energy levels tends to be proportional to $\dfrac{1}{\text{frequency of radiation}}$. 2

The ingenuity of humankind is often marvelled upon when such complicated devices as lasers and masers are used in devices such as DVD players and atomic clocks. However, nature itself is somewhat more modest – natural masers have been produced for billions of years in the atmospheres of stars, comets, star-forming regions, supernova remnants and even super massive black holes. And all this time the Universe has kept quiet about its technological achievements. 3

Essentially what happens in star masers is this – light and explosions from stars excite nearby gas regions. In these nearby gases, light and collisions lead to high energy levels becoming populated by electrons. Some of these higher energy levels will be metastable, setting up a maser amplifying region. As in any laser, the process has to start with a spontaneous emission accidently shooting off in the correct direction, but afterwards, stimulated emission takes over and the maser beam starts towards infinity; infinity is stretching the truth a bit but it does sound good. In reality the intensity of the maser beam increases rapidly and the beam propagates at the speed of light as em radiation tends to do. 4

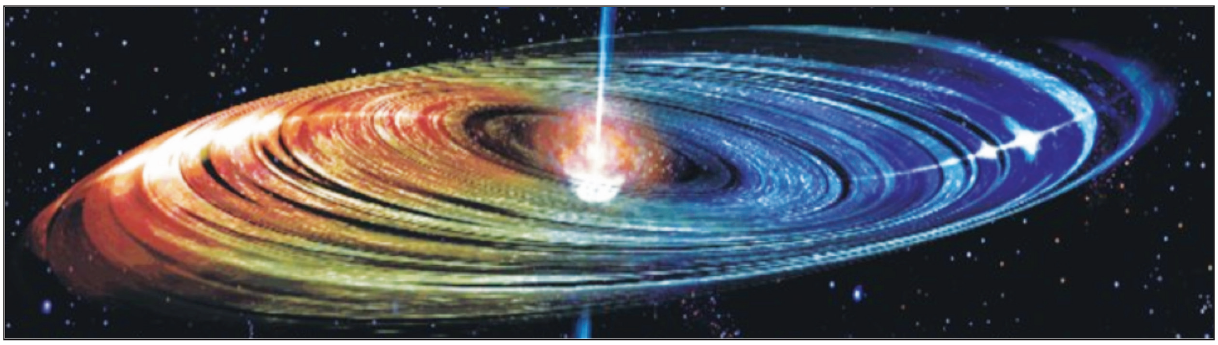

Diagram 1

In all fairness to humans, there is one aspect of laser design that nature has not succeeded in producing by itself and that is the resonant cavity of a laser. Multiple passes through a laser amplifying medium improves the quality of laser operation immensely. This is achieved among humans using two mirrors but as yet this technological advancement seems to be missing in gas regions around stars. 5

Unit 3 2017.pdf

Paragraph

Although a resonant cavity seems to be missing in natural masers there does seem to be a high degree of beaming. Small differences across the irregularly shaped maser cloud disk lead to vast differences in intensity due to exponential gain. The directions in the gas disk that have a longer length of population inversion will appear much brighter (as the increased maser amplification leads to an exponential increase). The majority of the radiation will emerge along this line of greatest length in a "beam"; this is termed *beaming*. Not quite as good as a laboratory laser cavity but pretty impressive for a seemingly random gas cloud. 6

Megamasers is the term used for water masers in the gas cloud around black holes. 7

Most large galaxies with a nucleus and a bulge have, at their centres, a supermassive black hole. When the black hole is actively accumulating matter, it releases a tremendous amount of energy, and the galaxy is said to have an active nucleus. Buffered by the dust in the flow of matter into the black hole, molecules can survive there and get energised by collisions with other molecules and dust particles. Water molecules are common in this environment, and they can emit maser radiation. A water maser can shine a beam of microwaves at a very specific frequency, 22.235 GHz, which corresponds to microwaves about 1 cm in wavelength. The primary reason megamasers have been studied in detail is that they make excellent tools to help us understand the environment around black holes. Small changes in the detected frequency of the masers, owing to the Doppler effect, mean that we can determine the line-of-sight velocity of maser clouds very precisely. All this means that the mass of black holes can be determined 20 times more accurately. 8

Another primary science goal that we can address with studies of water megamasers is measuring distances to galaxies. Measuring distances is a notoriously difficult problem in astronomy, and one of the most important. By analysing the internal dynamics of water maser systems, we can measure the rotation velocity, v, of maser clouds as they orbit the black hole from the Doppler shift of the maser lines. We can also measure the centripetal acceleration, a, of maser clouds by observing how the Doppler shift velocity changes over time. Using the simple relation for centripetal acceleration: 9

$$a = \frac{v^2}{r}$$

we can then calculate the radius of the gas 'disk' (see below) orbiting the black hole.

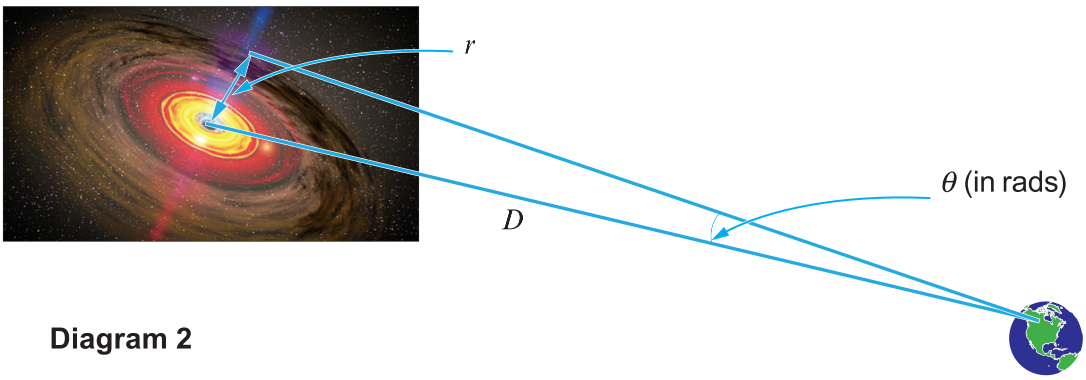

Diagram 2

We can also measure the angular size (θ) of the gas disk (in radians). Given the angular size (θ) and the radius, r, of the gas disk, we can obtain the distance, D by assuming that the angle (θ) is small. The simplicity of this method is remarkable and gives reliable results that are not dependent on controversial or unproven theories. In fact, the results obtained thus far from a couple of suitable galaxies with megamasers leads to a Hubble constant of (68.9 ± 7.1) km s^{-1} Mpc^{-1}. 10

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7. Answer the following questions in your own words. Direct quotes from the original article will not be awarded marks.

(a) In your own words explain what a megamaser is and how it works. (See paragraphs 1, 4, 7 and 8.) [5]

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(b) If the lifetime of a metastable level leading to visible light is $2\,\mu s$, estimate the lifetime of a 22.235 GHz maser transition. (See paragraph 2.) [3]

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(c) The recessional velocity of the galaxy used in paragraph 10 is 3 300 km s^{-1}. Use Hubble's law to calculate the distance of the galaxy in Mpc. (N.B. use the value of H_0 in paragraph 10. There is no need to change the units.) [2]

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(d) What does the author mean by "We can also measure the centripetal acceleration, a, of maser clouds by observing how the Doppler shift velocity changes over time"? (See paragraph 9.) [3]

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(e) (i) Doppler shift measurements are obtained for a gas disk orbiting a black hole. The orbital velocity of gas around the black hole is measured as 410 km s^{-1} and the acceleration of the gas disk is measured as 6 km s^{-1} **per year**. Calculate the distance of this region of the gas disk from the black hole. (See paragraph 9.) [3]

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(ii) The angular size (θ) in radians of the gas disk is measured as 5.1×10^{-9} rad $\pm$ 10% and its distance from Earth is measured as 1.53×10^{23} m $\pm$ 10%. Evaluate whether or not these values are consistent with your calculation in part *(e)*(i). (See Diagram 2.) [4]

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Unit 3 2018.pdf

SECTION B

Answer **all** *questions*

Read through the following article carefully.

Freely adapted from:

The Physics of Optical Fibres
By Justino Luis Moreno

Paragraph

Basically, an optical fibre is just a piece of glass along which you send light. A similar procedure was first demonstrated not with glass but with water flowing from a spout in 1840 by scientists Daniel Colladon and Jacques Babinet. Babinet later published his work in an article entitled "On the reflections of a ray of light inside a parabolic liquid stream". This effect can be reproduced quite easily in a school lab using the apparatus shown in Figure 1. 1

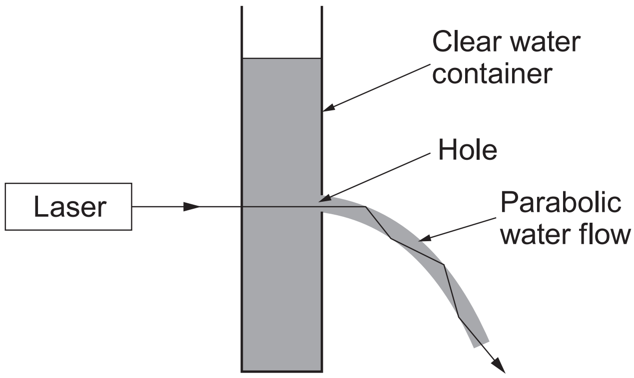

Figure 1

Although this set up is pleasing to the eye and is the basis of some water features, it isn't much use for international telecommunication! A standard optical fibre is shown in Figure 2 along with a ray of light entering it. 2

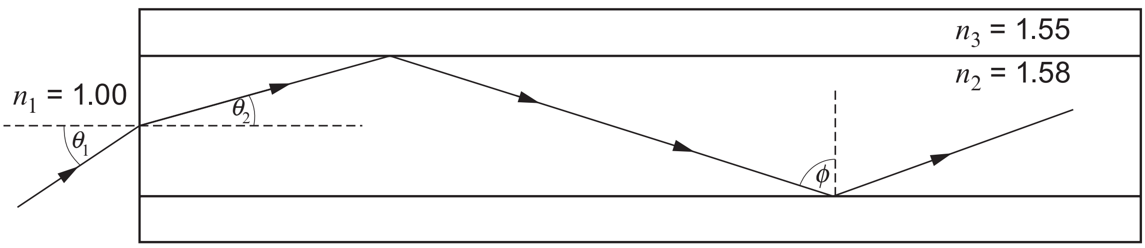

Figure 2

This ray of light is repeatedly reflected along the length of the optical fibre. If the entrance angle (θ_1) is small enough then an effect called total internal reflection (TIR) means that the light is completely reflected each time resulting in no light escaping as it travels along the optical fibre. The physicists in charge of designing these things can prove that the minimum value angle ϕ can have for TIR to take place is given by the equation: 3

$$\phi = \sin^{-1}\left(\frac{n_3}{n_2}\right)$$

Paragraph

This angle is around 80° for the optical fibre shown. The minimum angle for ϕ means that there is a maximum angle for θ_1. This gives an acceptance cone where the input light gets propagated without loss. A quick bit of geometry shows you that the exit angle is the same as the entry angle so that when light exits it produces an almost perfect cone shaped beam with a circular cross-section as can be seen in Figure 3. 4

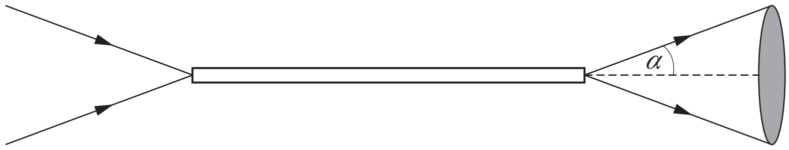

Figure 3

An important term in optical fibre technology is maximum bit rate. Since each pulse represents a bit of data, this is the highest frequency of pulses that can be sent down an optical fibre before the pulses start to overlap and become indistinguishable from each other. A monomode optical fibre cable of length 80 km can comfortably have a maximum bit rate of 10 Gb s^{-1}. This means that 10^{10} pulses can pass along its length every second without overlapping. A typical telephone conversation requires around 10 kb s^{-1} meaning that one monomode optical fibre cable can carry a million telephone conversations simultaneously. High definition TV requires a much higher bit rate and a 10 Gb s^{-1} fibre will only handle around 2 000 high definition TV signals. 5

One of the most important factors that limit data transfer in optical fibres is multimode dispersion. This, put simply, is to do with the entrance cone of light in Figure 3. There is a range of distances that the pulses have to travel because there are a variety of angles at which they can travel along the fibre. The pulses then become spread out and indistinct, ruining the digital signal. Multimode dispersion is eliminated by using monomode optical fibre cables which have very thin cores. As a rule, monomode fibres have a core diameter of around 8 μm. This means that the core is less than 10 wavelengths thick so that the light stops behaving like rays. For monomode fibres there is only one propagation direction – along the axis. 6

Another drawback of sending signals down long lengths of optical fibres is that some of the light is either scattered or absorbed by the glass molecules themselves (an effect known as attenuation). Although no light escapes the fibre due to TIR there are other losses involved and these are usually summarised by using a decibel (dB) scale. This scale is defined by saying that a 10 dB decrease (-10 dB) in power is when the power has dropped to 10% of its input value. A loss of -20 dB then corresponds to a drop to 1% power and -30 dB is a drop to 0.1% power. The following table shows the relationship between dB values and power ratio: 7

dB	Power ratio $\left(\frac{P}{P_0}\right)$
-5	0.316
-10	0.100
-15	0.032
-20	0.010

Table 1 8

The losses of optical fibres are usually quoted in the unit dB/km and some modern optical fibres can have values as low as 0.01 dB/km. Hence, each km of cable loses 0.01 dB meaning that you can use 1 000 km before your signal is down to 10% strength. Optical fibres have come a long way since they were born in a fountain of light nearly 200 years ago. They stretch out (literally) to all areas of the world bringing light, sound and broadband wherever they go. Technical advances mean that data can be sent at a rate of 1.05×10^{15} pulses per second over a distance of 50 km with only one monomode optical fibre. Nonetheless, the technology has its limitations of which attenuation and multimode dispersion are but two. 9

Unit 3 2018.pdf

8. Answer the following questions in your own words. Direct quotes from the original article will not be awarded marks.

(a) Explain why the water in Figure 1 flows in a downward curve (paragraph 1). [2]

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(b) Show that the minimum angle (ϕ) for total internal reflection in the optical fibre of Figure 2 is less than 80° (paragraph 3). [2]

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(c) Explain why the light emerging from the optical fibre is in the shape of a cone of light (paragraphs 3 & 4 and Figures 2 & 3). [2]

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(d) Calculate the angle (α) of the cone of light emerging from the optical fibre in Figure 3 (see paragraphs 3 & 4 and Figures 2 & 3). [4]

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(e) Rhys makes the following claim:

"A multimode optical fibre that can transfer data at a maximum bit rate of 500 Mb s^{-1} over a distance of 1 km would be able to transfer data at a maximum rate of 50 Mb s^{-1} over a distance of 10 km."

Discuss whether or not Rhys's claim is valid (paragraph 5 & 6). [3]

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(f) A certain type of optical fibre has an attenuation of 0.8 dB/km. When the signal decreases to 6% of its original intensity it must be amplified or the signal will be lost. An engineer intends to install these optical fibres in lengths of 20 km before each amplifier. Determine whether or not this is an appropriate length of optical fibre to use (see paragraph 7 and Table 1). [3]

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(g) Aled claims that a core diameter of around 13 μm is thin enough for a monomode optical fibre because a wavelength in air of 1.3 μm is usually used in communications. Rhian claims that this is nonsense because the wavelength in the optical fibre is changed due to its refractive index. Justify by including a calculation who is correct (paragraph 6). [4]

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END OF PAPER

Unit 3 2019.pdf

SECTION B

Answer **all** *questions*

Read the following article carefully.

A Brief History of Particle Physics

Freely adapted from http://www.particleadventure.org/other/history/quantumt.html

Paragraph

The story starts a long time ago (~400BC) in ancient Greece. Democritus, Leucippus and Epicurus developed the theory that *the universe consists of empty space and an (almost) infinite number of invisible particles* which differ from each other in *form, position, and arrangement.* All matter is made of indivisible particles called atoms. 1

It could be argued that very little happened for another 2000 years until in 1654 Otto von Guericke invented a vacuum pump. You will probably have heard about one use he made of this vacuum pump – the Magdeburg hemispheres. Two metal hemispheres with a vacuum between them could not be separated by teams of horses pulling them apart. 2

Otto von Guericke also invented an electrostatic generator – electron accelerators were born and lightning discharge could be produced artificially. 3

In 1705, it was noted that by combining both inventions of Otto von Guericke, lightning discharge can go between electrodes in a gas and can go further at low pressures. 4

In 1838, Michael Faraday noted a strange glow from low pressure gases in glass tubes when the gas was conducting electricity. Some 40 years later Eugen Goldstein noted so-called "cathode rays" in these gases. In 1898, Joseph (JJ) Thomson identified these cathode rays as streams of negatively charged particles and made measurements on the properties of these electrons. He then put forth his "plum-pudding" model of the atom. In this model, the atom is a slightly positive sphere with small, raisin-like negative electrons inside – he obviously didn't know that radioactivity, researched by Marie Curie, was a nuclear phenomenon. Hardly surprising since the nucleus hadn't been discovered yet…. 5

The early 20th century saw Hans Geiger and Ernest Marsden, under the supervision of Ernest Rutherford, scatter 4.7 MeV alpha particles off a gold foil and observe large angles of scattering. This put an end to the plum-pudding model and suggested that atoms have a small, dense, positively charged nucleus of diameter around 10^{-14} m surrounded by electrons orbiting at a distance of around 10^{-10} m. 6

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Unit 3 2019.pdf

Paragraph

Shortly afterwards, Niels Bohr succeeded in constructing a theory of atomic structure based
on quantum physics. Then, in 1924, Louis de Broglie proposed that matter had wave-like 7
properties.

Strangely enough, de Broglie's wave particle duality led to JJ Thomson's son (George Paget
Thomson) obtaining one of the very first electron diffraction patterns for crystals. Whereas his
dad obtained a Nobel prize for identifying the electron as a particle, he obtained the prize for 8
its wave properties.

The first evidence for a proton was found by Ernest Rutherford in 1919 and, in 1921, James
Chadwick and E.S. Bieler concluded that some strong force holds the nucleus together. 9

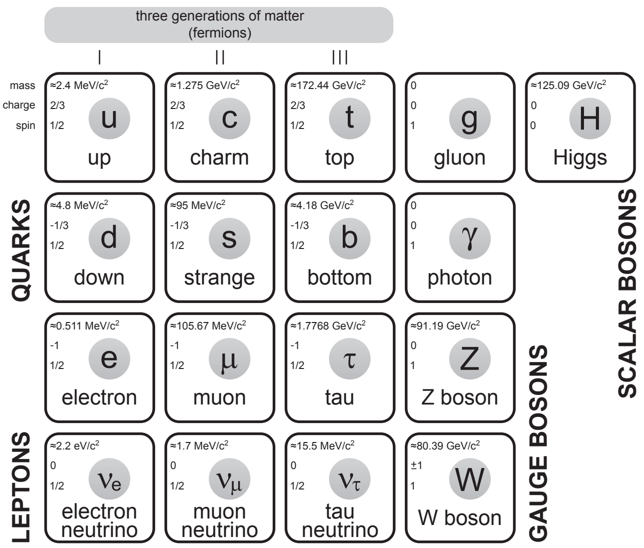
Paul Dirac realised that the positively charged particles required by his equation were new
objects (he called them "positrons"). They are exactly like electrons, but positively charged 10
– the first example of antiparticles. James Chadwick discovered the neutron. Both these
developments took place in 1931.

Enrico Fermi put forth a theory of beta decay that introduced the weak interaction. This is the
first theory to explicitly use neutrinos and particle flavour changes (1933-34). 11

In 1947, a meson that interacts strongly was found in cosmic rays, and was determined to be
the pion. A year later, the Berkeley synchro-cyclotron produced the first artificial pions. 12

Some 20 years later, in the late 1960s, electron scattering experiments were carried out using
the Stanford Linear Accelerator. High speed electrons were scattered off protons and the 13
electrons appeared to be bouncing off multiple small hard cores inside the proton. James
Bjorken and Richard Feynman analysed this data in terms of a model of constituent particles
inside the proton.

Standard Model of Elemental Particles



In a summary talk for a conference, John
Iliopoulos presented, for the first time in 14
a single report, the view of physics now
called the Standard Model – this was 1974.

After eighteen years of searching with
many accelerators, experiments at the
Fermilab near Chicago discovered the top 15
quark at the unexpected mass of 172 GeV.
No one understands why the mass is so
different from the other five quarks.

Almost half a century after Peter Higgs
predicted a Higgs boson as part of a
mechanism (invented by a number of 16
theorists) by which fundamental particles
gain mass, the ATLAS and CMS
experiments at the CERN lab discovered
the Higgs boson (2012).

Some two and a half millennia after the commencement of particle physics it seems we are
a lot closer to understanding the Universe but there are still many questions to be answered.
Some of the most fundamental questions to be answered are: What is dark matter? What is 17
dark energy and do either of these exist? How does quantum gravity work? For the answers to
all these questions, and any others, please consult your physics teacher.

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7. Answer the following questions in your own words. Extended quotes from the original article will not be awarded marks.

(a) Calculate the maximum force required to separate two Magdeburg hemispheres of radius 10 cm (atmospheric pressure = 1.01×10^5 Pa, see paragraph 2). [2]

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(b) Suggest why lightning discharge travels further at low pressures (see paragraphs 4 & 5). [2]

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(c) (i) Calculate the distance of nearest approach for a 4.7 MeV alpha particle when fired at gold foil (the charge on a gold nucleus is $+79e$, see paragraph 6). [3]

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(ii) State why your answer to *(c)*(i) and the results of the alpha particle scattering experiment contradict the “plum-pudding” model (see paragraphs 5 & 6). [2]

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Examiner only

(d) A student claims that a 4 keV electron is suitable for atomic diffraction experiments. Determine whether or not she is correct (see paragraphs 6 & 8). [4]

(e) Explain why the strong nuclear force is required to hold nuclei together (see paragraphs 6 & 9). [1]

(f) A typical thermal electron has a rms speed of approximately 100 000 $m\,s^{-1}$. When a positron annihilates an electron at room temperature, two photons of wavelength 2.43×10^{-12} m are produced. Explain, using conservation of momentum, why these photons travel in opposite directions (see paragraph 10). [3]

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(g) Explain why the quark make up of a π^- meson is $\bar{u}d$. [1]

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(h) Calculate the mass of a top quark in kg (see paragraph 15).

Note that $1\,u \equiv 931$ MeV if required. [2]

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END OF PAPER

Unit 3 2022.pdf

SECTION B

Answer **all** questions.

6. Read through the following article carefully.

How do you produce really low temperatures and can you then do some cool stuff with it?

Paragraph

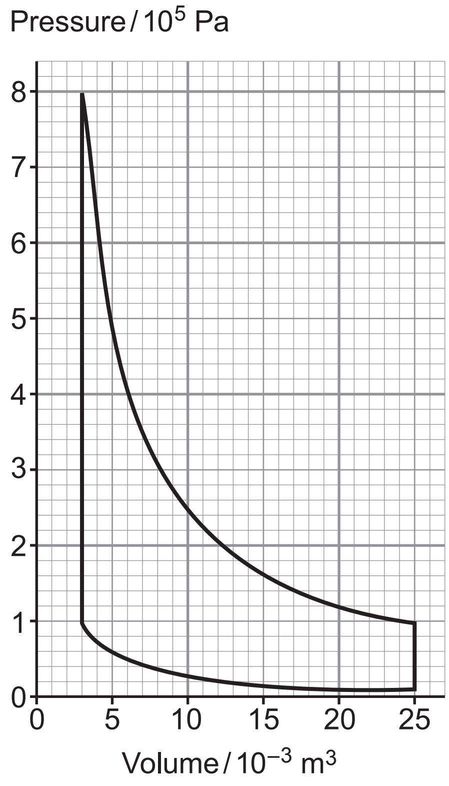


Figure 1

The first thing you have to do is to produce low
temperatures and this is usually achieved by 1
taking a gas through a closed cycle such as that
shown in Figure 1.

1. Air, initially under high pressure, is expanded at constant temperature.
2. Heat is allowed to escape at constant volume. 2
3. The gas is compressed at a constant low temperature.
4. The gas is heated at constant volume to a high pressure.

The end result is that the gas does a large amount
of work in each cycle. This means that the gas 3
must have heat flowing into it leading to the
cooling of whatever it is that requires cooling.

A different and simpler method of cooling gases is called the Joule-Thompson method,
which does not involve a gas going through a cycle. A gas is compressed to a pressure of
two thousand times atmospheric pressure. The gas is then allowed to cool back to room
temperature. After that, it is allowed to expand quickly through a nozzle to atmospheric 4
pressure leading to a tremendous amount of further cooling. In the expansion, the gas does a
very large amount of work but the non-ideal gas molecules also have an enormous increase in
potential energy because of the attractive forces between molecules. Both these factors cool
the gas by a huge factor.

Okay, so now we have the basic physics for producing low temperatures. Apply this low
temperature to air and we can make a product that can give hours of fun – liquid nitrogen 5
(LN_2).

Most of the fun that can be obtained from LN_2 is based on the fact that it boils at a temperature
of –195.8 °C and, in doing so, expands by a factor of approximately 1000. For example, a cool 6
rotating rocket can be built using a simple plastic water bottle.

Unit 3 2022.pdf

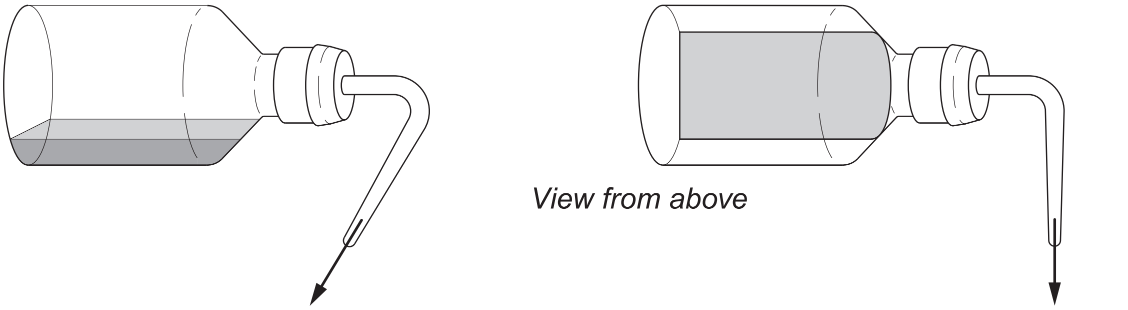


Figure 2

Paragraph

The boiling nitrogen makes the nozzle of the plastic water bottle into a jet engine. When you then place the plastic water bottle on a nicely polished wooden floor it will rotate far more quickly than you would ever anticipate, providing a great demonstration for even the most ardent physics-hating pupil. What can be even more spectacular is when the bottle accidentally hits something solid (like the leg of a lab desk). The wall of the plastic water bottle will be at around –200 °C and fragile. This, combined with the high pressure inside the bottle, leads to rather a loud and impressive bang. 7

Although LN_2 can be fun, it can also be dangerous. For instance, when transporting a large, open container of LN_2 (see Figure 3), you should not remain in the confined space of a lift with the container in case the lift becomes stuck. Most of the danger from LN_2 arises from its 200 °C temperature difference from its surroundings. Nonetheless, it is always amusing when TV chefs wear safety goggles and gloves when making ice cream using LN_2 but never bother with any safety equipment when using litres of boiling fat above a gas fire. I promise you that the boiling fat is far more dangerous even though the temperature difference, compared with human body temperature, is around 200 °C for both LN_2 and boiling fat. 8

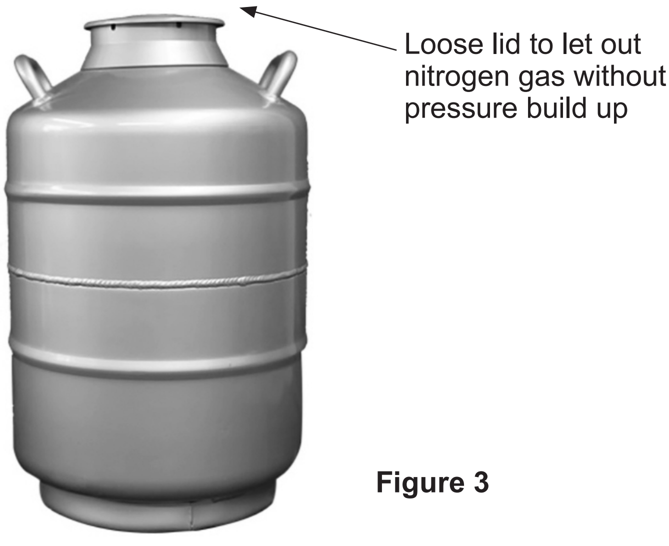


Figure 3

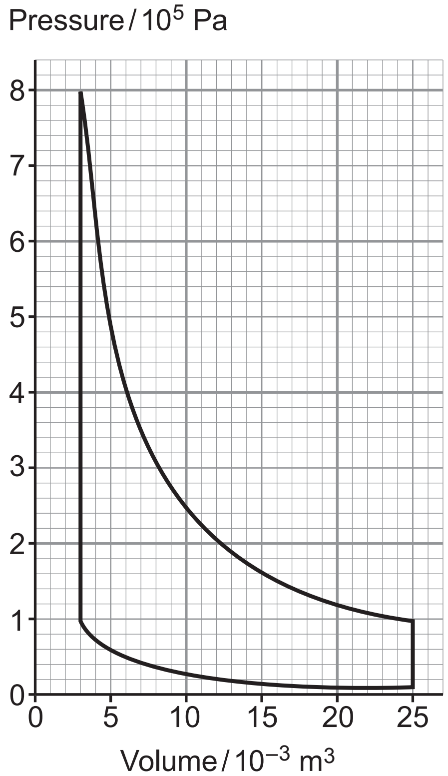
In conclusion, low temperatures and LN_2 can be dangerous for some obvious reasons but they can also lead to some amusement and even some Michelin-starred ice cream. 9

Unit 3 2022.pdf

Examiner only

Answer the following questions in your own words. Direct quotes from the original article will not be awarded marks.

(a) **Label the diagram** with the processes 1–4 along with an arrow to show the direction of the cycle (see paragraph 2). [2]



(b) The author states that work is done by the gas in the cycle. By considering the four stages of the cycle, explain whether or not this statement is correct (see paragraph 3 and Figure 1). [2]

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Examiner only

(c) Estimate the work done during the cycle. [3]

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(d) When a gas is compressed in a metal cylinder it will become hot. Explain why the gas will then cool to room temperature (see paragraph 4). [2]

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(e) Explain whether or not the author is correct to state that the gas expanding quickly through the nozzle will cool greatly (see paragraph 4). [4]

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(f) Explain very briefly why nitrogen gas escapes through the nozzle of the plastic water bottle (see Figure 2 and paragraph 7). [1]

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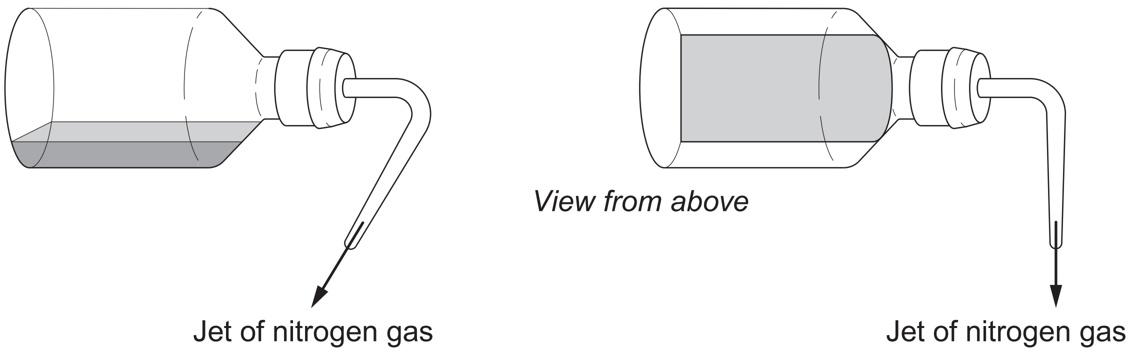
(g) Starting from Newton's 3rd Law, explain why the plastic water bottle shown will rotate anti-clockwise. [2]



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(h) An exploding bottle produces a "loud bang" (see paragraph 7). Explain briefly why this "loud bang" can be heard more than 100 m from the exploding bottle. [2]

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(i) Justify the author's statement that it would be dangerous to share a confined space (like a lift) with a large container of liquid nitrogen (see paragraph 8 and Figure 3). [2]

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END OF PAPER

Unit 3 2023.pdf

SECTION B

Answer **all** questions.

7. Read through the following article carefully. Paragraph

A summary of binary stars

First of all, we call two stars orbiting their common centre of mass a "binary star". Binary stars fall into 5 different classes, categorised by the detection method: 1

1. Visual binaries
2. Spectroscopic binaries
3. Eclipsing binaries
4. Astrometric binaries
5. Non-eclipsing binaries that can be detected using photometry.

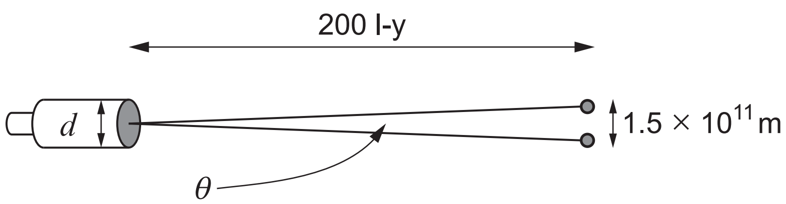
Visual binaries

These do what it says on the tin – the angular separation of the stars is great enough that both stars can be seen as separate objects through a telescope. There is a simple relationship between the minimum angular separation θ (in radians), the diameter, d, of the telescope and the wavelength, λ, of the radiation used. 2

$$\theta = 1.22\frac{\lambda}{d} \quad \text{Equation 1}$$

This equation means that a nearby binary star that is around 200 light years (l-y) away will only be visible as two stars with a d = 8.0 m telescope if the two stars are separated by at least the Earth-Sun distance of 1.5×10^{11} m (see Figure 1). 3

Figure 1



Perhaps the most famous of visual binary stars is the brightest star in the night sky – Sirius. Sirius is not, in fact, a single star but a binary consisting of Sirius A and Sirius B that orbit each other with a period of around 50 years. Sirius A is a main sequence star which has roughly 2 times the mass of the Sun, 1.7 times the Sun's temperature and 25 times its luminosity. Sirius B is a white dwarf with a similar mass to our Sun, a surface temperature 4.3 times greater than our Sun but a luminosity 18 times less than our Sun. 4

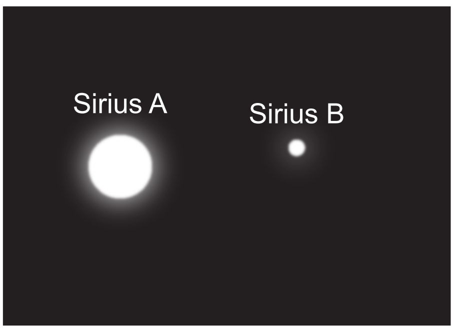


Figure 2

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Spectroscopic binaries

These are binaries where the stars cannot be viewed separately but the existence of the
two stars can be inferred from the variation of the Doppler shifted light from the system.
Both stars will orbit their mutual centre of mass and hence their Doppler shifts will vary 5
periodically during their orbits. This theory is the same as that used to discover exoplanets
from the wobble of their parent star.

Eclipsing binaries

These happen when the line of sight of the observer is in the plane of the orbit of the two
stars. When this happens, one star can pass directly in front of the other leading to a drop 6
in the intensity of light entering the observer's telescope. An idealised graph of intensity
against time is shown in Figure 3.

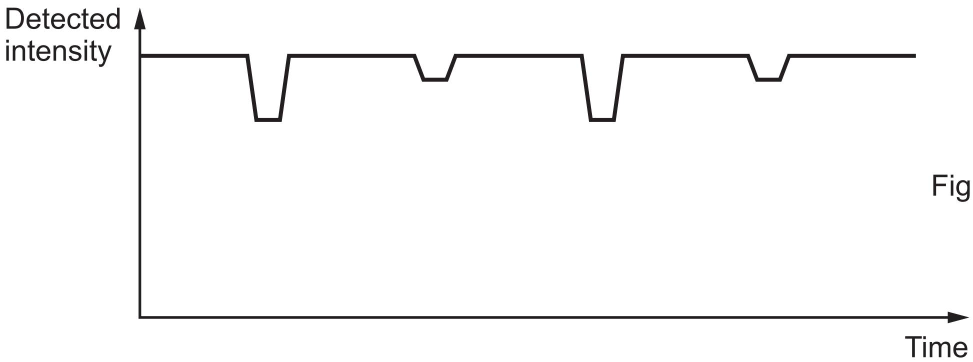

Figure 3

In general, there are two eclipses for each orbit. The larger of the two dips in intensity 7
occurs when the hotter of the two stars is blocked by the colder star.

Astrometric binaries

These are binary systems where only one star is visible and that star can be seen moving
in an orbit around something which is invisible. These are different from spectroscopic
binaries because the visible star can actually be seen to move rather than inferring its 8
motion from the variation of the Doppler shift. Stars orbiting neutron stars have been
observed by this method but the ultimate goal of this technique would be to observe a star
orbiting a black hole.

Non-eclipsing binaries that can be detected using photometry

This simply means detecting binaries by measuring light intensity but not via the eclipsing
method described earlier. It turns out that there are three different techniques of finding
binaries within this class of binary. First, an increased light intensity can be detected
periodically from the binary system – this is caused by light being reflected from one
star, off the other and into the Earth's telescope. Second, one star can exert such a 9
strong gravitational pull on the other that it becomes deformed – this leads to a periodic
fluctuation in the light intensity detected. Third, is an effect sometimes known as Doppler
beaming – a star moving towards an observer will lead to an increased number of photons
detected per second (as well as a decrease in the wavelength).

These 5 different classes of binary stars can provide an enormous amount of valuable
data regarding stellar orbits and masses but they can also allow measurement of galactic 10
distances and even give insights into black holes and supernovae.

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Examiner only

Answer the following questions in your own words. Direct quotes from the original article will not be awarded marks.

(a) A light year is 9.46×10^{15} m. Use equation 1 and a wavelength of 500 nm to show that the author's claim about an 8 m telescope viewing the binary star mentioned in paragraph 3 and Figure 1 is correct. [4]

(b) (i) Show that the distance between Sirius A and Sirius B is approximately 3×10^{12} m (see paragraph 4). [Mass of the Sun is 2×10^{30} kg.] [2]

(ii) Show that the centre of mass of the Sirius binary system is located approximately 1×10^{12} m from Sirius A. [2]

Examiner only

(iii) Use data in paragraph 4 and Stefan's law to calculate the ratio of the radii of Sirius A and Sirius B: [4]

$$\frac{R_{\text{Sirius A}}}{R_{\text{Sirius B}}}$$

(c) (i) Evaluate whether the variation of light intensity shown in Figure 3 is consistent with an eclipsing binary system (see paragraph 6). [3]

(ii) Use Stefan's law to explain why the "larger of the two dips in intensity occurs when the hotter of the two stars is blocked by the colder star" (see paragraphs 6 and 7 and Figure 3). [2]

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(d) Explain why a star orbiting a much more massive black hole is an astrometric binary (see paragraph 8). [1]

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(e) One of the three methods for detecting non-eclipsing binaries that can be detected by photometry, suggests that stars do not behave as ideal black bodies. Explain which **one** of the three methods relies on stars not behaving as black bodies (see paragraph 9). [2]

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END OF PAPER

SECTION B

Answer **all** questions.

8. Read through the following article carefully. Paragraph

Some Interesting Aspects of Quantum Physics

1 The whole idea of quantum physics usually starts with the idea that the energy of photons is quantised. Following the work of Planck and Einstein, the smallest packet of electromagnetic energy (the photon) has an energy given by:

$$E = hf \quad \textbf{Equation 1}$$

2 This simple little equation is no stranger to A level physics students but within six years of its inception it had solved the problems of black body radiation (by not allowing too many high energy photons) and the photoelectric effect (by explaining why the kinetic energy of photoelectrons depends on the wavelength of light and not its intensity).

3 In 1913, quantum physics took another enormous leap when Niels Bohr applied it to electron orbitals in atoms. In his new theory, the energy levels of electrons are quantised and electromagnetic radiation is emitted or absorbed when electrons make jumps from one quantised level to another. In a later interpretation of the Bohr theory, the circumference of any electron orbit must be a whole number of electron wavelengths (so that the electron's de Broglie wavelength leads to a stationary wave).

$$n\frac{h}{p} = 2\pi r \quad \rightarrow \quad r = \frac{nh}{2\pi p} \quad \textbf{Equation 2}$$

4 where n is an integer, p is the momentum of the electron and r the radius of its orbit. If we are concerned with the hydrogen atom, equating the electrostatic force between the proton and electron with the centripetal force leads to Equation 3 (m_e and e are the mass and charge of an electron and ε_0 the permittivity of free space).

$$\frac{m_e v^2}{r} = \frac{e^2}{4\pi\varepsilon_0 r^2} \quad \rightarrow \quad p^2 = \frac{m_e e^2}{4\pi\varepsilon_0 r} \quad \textbf{Equation 3}$$

5 Substituting for r in Equation 3 from Equation 2 gives:

$$p = \frac{m_e e^2}{2\varepsilon_0 nh} \quad \textbf{Equation 4}$$

and the starting point of Equation 3 can also be used to show that the kinetic energy of the electron in its orbit is given by:

$$\frac{1}{2}m_e v^2 = \frac{e^2}{8\pi\varepsilon_0 r} \quad \textbf{Equation 5}$$

However, the electrostatic potential energy of the electron and proton in hydrogen is: 6

$$\mathrm{PE} = -\frac{e^2}{4\pi\varepsilon_0 r} \quad \textbf{Equation 6}$$

so that the total energy of the hydrogen atom is:

$$\text{Total energy} = \mathrm{PE} + \mathrm{KE} = -\frac{e^2}{4\pi\varepsilon_0 r} + \frac{e^2}{8\pi\varepsilon_0 r} = -\frac{e^2}{8\pi\varepsilon_0 r} \quad \textbf{Equation 7}$$

Using Equations 4 and 7 and the rather useful relationship: 7

$$\mathrm{KE} = \frac{p^2}{2m} \quad \textbf{Equation 8}$$

it is reasonably straightforward to show that the total energy of the hydrogen atom is:

$$\text{Total energy (in eV)} = -\frac{p^2}{2m} = -\frac{m_e e^3}{8{\varepsilon_0}^2 n^2 h^2} \quad \textbf{Equation 9}$$

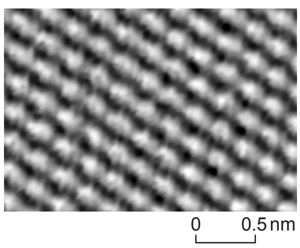
It turns out that when the integer n = 1, we have the ground state of hydrogen and n = 2
corresponds to the first excited state etc. By inputting n = 1 into Equation 9, we find that 8
the energy of the ground state of hydrogen is approximately -13.6 eV. Hence, the total
energy of the n^{th} level of hydrogen (in eV) is given by Equation 10.

$$\text{Total energy (in eV)} = -\frac{13.6}{n^2} \quad \textbf{Equation 10}$$

The success and accuracy of Niels Bohr's theory of the hydrogen atom was almost
unbelievable and meant that quantum physics was well and truly established as the main 9
research topic in science for the next century.

When, in 1926, Erwin Schrödinger developed the quantum mechanical wave equation,
there was another explosion in quantum physics research. Schrödinger's publication of 10
1926 included a complete (non-relativistic) solution of the hydrogen atom which is the
foundation of the whole subject of chemistry!

Another highlight directly obtainable from his famous wave equation is the concept of
quantum mechanical tunnelling – where particles can tunnel through energy barriers
that are too high for them. This is the explanation for alpha particles escaping from large
nuclei in the process of alpha decay. The strong nuclear force is so great that the energy 11
required for an alpha particle to escape a nucleus is far too big and
no alpha particles should ever be detected. However, this tunnelling
effect means that there is a small but finite chance that alpha
particles can, indeed, escape.



This tunnelling effect has also led to microscopes that can "see"
individual atoms e.g. the graphite atoms in the hexagonal pattern in 12
the image.

Examiner only

Answer the following questions in your own words. Direct quotes from the original article will not be awarded marks.

(a) (i) Explain how Einstein's photoelectric equation arises from Equation 1 ($E = hf$) and the principle of conservation of energy (see Paragraphs 1 and 2). [2]

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(ii) The intensity of light does not affect the maximum kinetic energy of photoelectrons emitted from the cathode of a photocell. State what electrical quantity is proportional to the intensity. [1]

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(b) Show how the substitution for r in Equation 3 $\left(p^2 = \frac{m_e e^2}{4\pi\varepsilon_0 r}\right)$ using Equation 2 $\left(r = \frac{nh}{2\pi p}\right)$ gives Equation 4 $\left(p = \frac{m_e e^2}{2\varepsilon_0 nh}\right)$. [2]

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(c) (i) Use Equation 9 $\left(\text{Total energy (in eV)} = -\dfrac{m_e e^3}{8\varepsilon_0{}^2 n^2 h^2}\right)$ to show that the energy of the n = 1 state of hydrogen is −13.6 eV. [2]

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(ii) Use Equation 10 $\left(\text{Total energy (in eV)} = -\dfrac{13.6}{n^2}\right)$ to explain why the ionisation energy of hydrogen is 13.6 eV. [2]

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(d) A student suggests that the hydrogen 656 nm red line is emitted from a hydrogen atom when an electron drops from the second excited level (n = 3) to the first excited level (n = 2). Determine whether or not this is true.

Use Equation 10 $\left(\text{Total energy (in eV)} = -\dfrac{13.6}{n^2}\right)$. [3]

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Examiner only

(e) Explain how quantum tunnelling will provide a lower than expected value for V (see Equation 11 $\left(eV = \frac{hc}{\lambda}\right)$ and Paragraph 14). [2]

(f) (i) Explain why an increased temperature provides a greater current in a LED (see Paragraph 15). [2]

(ii) Calculate a typical thermal kinetic energy of a particle at room temperature in eV and use this to explain why it is difficult to measure the "switch-on" voltage of a diode accurately. [4]

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END OF PAPER